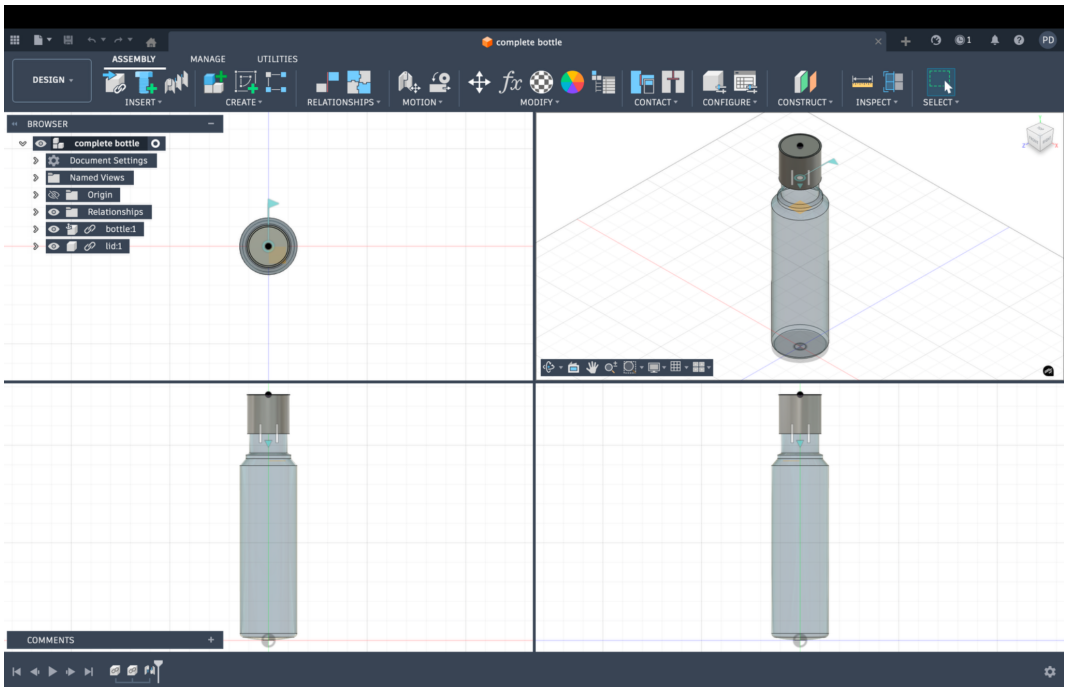
AutoCAD 360 / Fusion 360 – 3D Product Modeling & Animation



This project represents my hands-on learning journey with **AutoCAD 360 (Fusion 360)**, where I explored the fundamentals of **3D product design, parametric modeling, and basic animation** through a practical, real-world object. The primary outcome of this exercise was the complete design and visualization of a **1-litre water bottle with a functional lid assembly**, focusing on both form and functionality.

I worked extensively with core modeling tools such as **Extrude, Loft, Revolve, Shell, Fillet, and Combine** to build accurate and clean geometry. Special attention was given to maintaining **uniform wall thickness** using the Shell tool, replicating realistic manufacturing constraints. The bottle body and lid were modeled as separate components and later assembled, helping me understand **component hierarchy, alignment, and design intent** within an assembly environment.

To enhance realism, I applied **material properties** and simulated a **water-filled interior**, which improved my understanding of visual representation and material behavior in product design. I also explored **motion and animation tools** by creating a simple **lid opening and closing animation**, allowing me to visualize mechanical movement and interaction between parts.

Throughout this project, I developed a stronger grasp of **design workflow, sketch-based modeling, feature-based construction, and iterative refinement**. This exercise not only improved my technical proficiency in AutoCAD 360 but also strengthened my ability to translate everyday objects into **functional digital prototypes**, preparing me for more advanced product design and manufacturing-oriented projects.